

there are corner cases where products fail despite being used correctly.

Let us take the case of a wearable that has a magnetic-field-sensitive sensor. In the normal usage, it does not get impacted, but if a user wearing the device goes near an equipment that has a high magnetic field (like a big transformer), the wearable can fail due to the sensor being impacted by the external field. A disclaimer cannot adequately describe this situation, and the user may not be able to avoid going near the system. In these cases, product shielding needs to be designed so that no external magnetic field can impact the wearable device.

Second issue is the biocompatibility of the material used. Since the wearable device will be in contact with users' skin for extended periods, materials used should be safe and not cause any skin problems. This poses challenges in terms of costs and processes. Biocompatible materials are sometimes expensive and also need special processes to manufacture. Designers need to factor this in the design phase itself. This problem is acute when the product enclosure is metallic.

In addition, if devices have probes that are metallic and need to touch the skin, these invariably need special gels to ensure good contact, and the materials also need to be safe for use.

One of the frequently faced problems is water ingress inside the wearable devices. In many cases, initially, the product remains waterproof but sometimes seals that were used for prevention of water ingress wear out over a period of time, in turn, losing the ability to prevent the water from entering. Designers need to factor this as well and ensure that proper material is selected.

Designers can define the product usage period (useful life) so it can be discarded once the prescribed time is over. This works well when the product cost is low, but if the product is expensive, a proper mechanism of



Fig. 4: A conventional test pad and a bead probe test pad

inspection and preventive check is a must if the design has aging issues. This is especially true in the case of a wearable device used for medical applications, where the product cost is high and customers tend to use it for a longer period.

Finally, sometimes wearable devices may have consumables such as gels, pads and wipes. These have to be consistent and their tolerances have to be within the specified limits. Many a time, as time passes, cost pressure takes precedence and companies switch over to a low-cost material, which impacts the product, leading to failures. This happens also when some critical inputs are forgotten or missed out. This must be kept in mind.

Support phase. Most wearable devices are use-and-throw types at the end of their life or when gone faulty. This is due to the low cost of the products and high repair costs. Product manufacturing process in this case uses a specialised procedure that cannot be replicated in the repair centre.

However, when the product is expensive (like some medical equipment), it must be supported and would need to be repaired. If this is a requirement from the beginning, designers have to plan and implement this in the design phase itself. They need to clearly define a process, which will achieve the same type of assembly that is done at the manufacturing line, ensuring product integrity and reliability.

An important support issue is software upgrade; most wearable devices when launched do not have the full complement of the features and may also have bugs. Designers need to make provisions for the following two important aspects:

1. Provide a reliable communication mechanism through which the software can be upgraded.

2. Provide adequate internal memory so that if the upgrade process fails, the product can still work with an earlier version of the software.

Both of these provisions should be planned in the design phase and implemented.

Another way to tackle this issue is to keep the software functionality to bare minimum and carry out all complex operations either on the central application or on a smartphone, depending on where the application is running. This also ensures that cost of hardware is kept low and the software is simple, so that it can be tested and does not need an upgrade, if properly tested. With the option of software upgrade being done in the application, large-volume device software upgrade related issues can be reduced.

Wearable system software

Software for wearable devices is completely dependent on design consideration, functionality of the device and controllers used. However, there are certain generic elements that software developers need to keep in mind when deciding the software architecture of wearable devices. Following guidelines should be helpful when deciding on software architecture:

1. Use a standard-platform based architecture so that maintaining the software is easy.

2. Even when a simple two to three tasks functionality is needed, use of a real-time scheduler (kernel) allows predictable performance and ensures consistency in data, when data needs to be collected for a longer duration and data volume is high, along with time stamp. This particular aspect is very successful in a wearable device design, especially when healthcare functionality is implemented.

3. For complex functionalities, a Linux based platform is advanta-